

# Building Classification Using a YOLO Object Detection Model Trained on Aerial Imagery

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**Abstract**—Travel forecasting models rely heavily on land use data from government agencies, which can be inconsistent or unavailable in many regions. This study investigates the viability of aerial imagery and deep learning models as a first step to develop a method to obtain this statistical data.

Using Logan, Utah as a primary case study, the Microsoft Building Footprints dataset and aerial imagery was used to train a “You Only Look Once” (YOLO) model to classify building types, including residential, commercial, and apartment buildings. The model demonstrated high accuracy in identifying building types in the training city. However, when validated against five other Utah cities selected for diversity of median income and housing ages, the model showed limited transferability, which suggests that differences between house architecture necessitates local training.

This paper analyzes these performance discrepancies and discusses future potential research such as integrating additional environmental indicators and using high-resolution drone imagery to create more robust models for estimating household statistics and land use data.

**Index Terms**—aerial imagery; deep learning; AI; transportation; land use data

## I. INTRODUCTION

Travel forecasting models are based heavily on data provided by government statistical agencies, such as the US Census Bureau and the Bureau of Labor Statistics. These agencies provide the land use data needed to construct these models, giving statistics such as household number, median household income, etc. for census areas. In countries where statistical agencies are not as reliable, travel forecasting models are generally not developed or have data gaps. In countries where they are available, these models are highly reliant on the public release of such statistical data, which leads to problems if such surveys are delayed or canceled.

Recent research has sought to find alternative methods to collect land use characteristics. Gathering, analyzing, and utilizing remote sensing data from satellites has been a major focus over the last several decades, due to its ease of access to the public. Satellite aerial imagery has been used to analyze the volume and density man-made structures and roads to estimate area populations, in both urban and rural settings [Allaw et al. [1]][2]. Day time imagery has also been used to track vehicle movement between major cities as an indication of economic activity [3]. It has also been used to classify land use by estimating landscape patterns and building functions,

dividing different surfaces, and then categorizing building types [4]. Night time light data has also been shown to be useful as well. One study correlated city light output with social economic variables such as population, GDP and eclectic power consumption [5], while another used the light emitted from cars at night on highways to predict vehicle travel between cities [6]. Satellite imagery has also be used to produce data sets such as the Microsoft Building Footprints data set, which has been used along with LiDAR and point of interest data to inventory residential structures from census data [Microsoft [7]][8]. Unfortunately, processes and methods reliant on satellite data are also susceptible to the availability of said data, which like statistical data may not be available across the world or could have its public access revoked.

With the rise of commercially available and affordable aerial drones, the ability to capture remote sensing data such as high-quality aerial imagery has become significantly easier. Over the past several decades this has led to high-resolution and accurate products that has benefited many fields [9]. Drones have also seen greater application within the scientific sector [10]. The creation of open-source software’s for drone use has also increased its accessibility [11]. While not being as reliant on the public release of data, the practicality of and economic feasibility of wide spread use of drones is still being discussed, with ethical concerns of collecting higher resolution data still part of an ongoing discussion [12].

More recently, with the advancement of more accessible deep learning models, machine learning has become very prevent in analyzing and classifying remote sensing data. One of its primary uses so far has been to analyze large amounts of data, for such uses of tracking vehicle movement between cities [3], or predicting population characterizes and locations of man-made structures across entire countries [2]. Building classification is of particular interest for collecting land use characterizes. Models such as “You Only Look Once” (YOLO) are being used to identify buildings and extract building footprints [Nurkarim and Wijayanto [13]][14]. It is expected that as data sets continue to grow larger the use of deep learning models for analysis will become more and more relevant and practical, along with their potential to help solve a multitude of issues facing communities.

Within the context of this background and research the question arises “can remote sensing data be used to develop

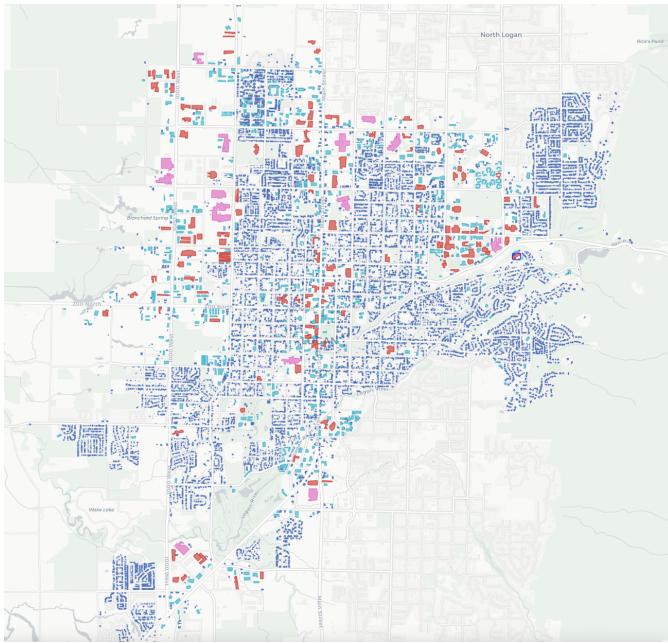


Figure 1: K-mean analysis for Logan, Utah

land use data currently obtained from statistical agencies?” The development of a process or method that could synthesize such data would solve many problems in places without reliable statistical data as well as supplement data in other places. With widely available deep learning models and aerial imagery, this process has become feasible to develop.

In this paper the process of collecting building footprints for a city, annotating aerial images of buildings, and training a YOLO model to classify building types is discussed. At the end it presents research and findings into creating a model that recognizes and categorizes building types based off remote sensing data, and discusses future research into estimating household statistics for a general area.

## II. METHODOLOGY

### A. Data Collection

The city of Logan, Utah was chosen for initial analysis. Logan was chosen for its size and variability of buildings ages and types. Building footprints for Logan were obtained from the Microsoft Building Footprint public dataset. This dataset allowed for easy access to building footprint data for this initial study. All analysis was run using Python 3.12 and various python packages and APIs.

With the chosen city the latitude and longitude of the official city boundaries was obtained. Using this boundary, all the building footprints in the city were extracted from the Microsoft Building Footprint dataset and saved. The buildings were then classified using a simple K-mean analysis to differentiate different sized buildings. To do this, the optimal K-mean was found using the elbow method. Using this K value, a K-mean analysis was run comparing the areas of the building footprints to their perimeters. Figure 1 shows the distribution



Figure 2: YOLO prediction example

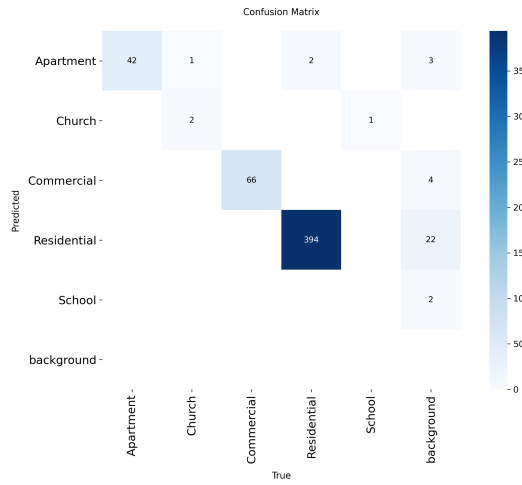
of different cluster types, with most of the buildings in Logan being a smaller size.

Using the building footprint location and the python Library “Contextily”, 600 randomly chosen aerial images of building in Logan were taken from publicly available satellite imagery and cropped to focus on a single building. 200 of the images were from clusters that included only larger buildings to help vary building types in the dataset.

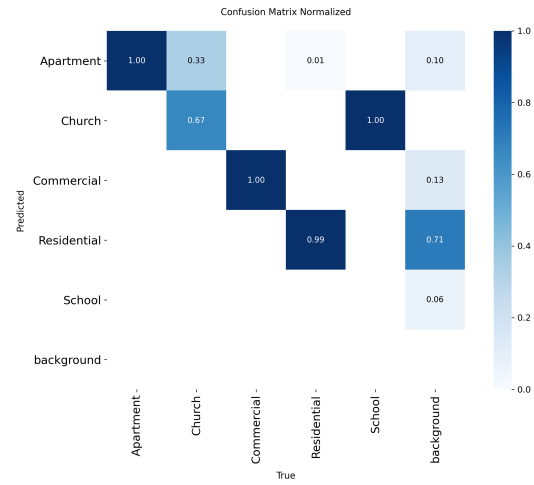
### B. Model Training

Ultralytics’ “You Only Look Once” or YOLO model is an open-source object detection model that has become very popular within the research community due to its accuracy and ease of use. The model is trained on images that have been manually annotated with bounding boxes, to identify the categories desired for the model to detect. The model trains itself on these images over hundreds of iterations until it can consistently recognize the different categories of objects. The model validates itself on other specified annotated images to ensure accuracy. The YOLO model is then able to predict these categories in nonannotated images by bounding them in bounding boxes and displaying the probability of correct identification. An example of a YOLO model prediction in Logan, Utah is shown in Figure 2.

Using the open-source tool Label Studio, the collected street map images were manually annotated. This annotation including creating bounding boxes around the roof of each building to categorize the building. The main buildings in each picture were labeled into five building categories: Apartment, Residential, Commercial, Church, and School. This order was constant across annotations. Best judgement was used in the annotation of the buildings, based off the appearance of the building and surrounding buildings and area. These bounding boxes were saved as text files and each one matched up with its corresponding image.



(a) Logan confusion matrix



(b) Logan confusion Matrix normalized

Figure 3: YOLO confusion matrixes for Logan, Utah.

The annotated data sets were split into two categories. 80% of the images were used in training the object detection model, while 20% of the images were reserved for validation purposes. Within these image sets clusters that included larger and less common building sizes were specifically included, due to their relatively lower likelihood of appearing, to make sure the model would be trained on these less common building types. Despite this, building types such as schools and church buildings were still underrepresented and only appeared a hand full of times, due to their relatively low number even within larger buildings.

The model was run through hundreds of iterations until it was able to accurately and reliably predict building types in Logan. The model was then validated on images it had never seen to help prevent the model from “memorizing” the images, ensuring accurate results. The results of the validation of the model are shown in Figure 3.

### C. Model Validation and Comparison

Once the model was trained and validated using the Logan, Utah data set, the model was then tested upon other cities in Utah to assess its viability in predicting building categories elsewhere in Utah. Using American Community Survey data, five other test cities were chosen based off the city’s median income level and the median age of the homes. These factors were theorized to have a significant impact on the architecture of the house and therefore the YOLO model’s ability to recognize and categorize them. The chosen cities can be seen in Table Ia.

Following the same method as previously described, building footprint data was collected and analyzed for each city. 200 aerial images of buildings were randomly chosen and annotated. These annotated data sets were then run through the model using validation only. The results of these runs are given below in Figure 4. These results were then analyses to

Table I

(a) Chosen Cities Based off Median Income and House Age

| House Age  | High Income      | Middle Income    | Low Income  |
|------------|------------------|------------------|-------------|
| Old Houses | Bountiful, Utah  | Ogden, Utah      | Price, Utah |
| New Houses | Saratoga Springs | St. George, Utah | (None)      |

evaluate the ability of the model to predict in other cities, as well as assess if household income or house age were indicating factors.

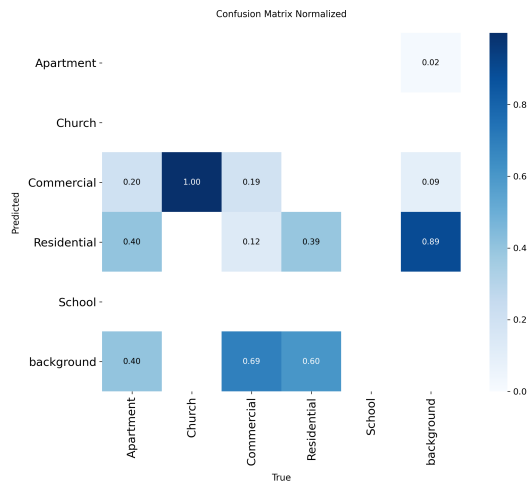
## III. RESULTS AND DISCUSSION

### A. Logan, Utah Results

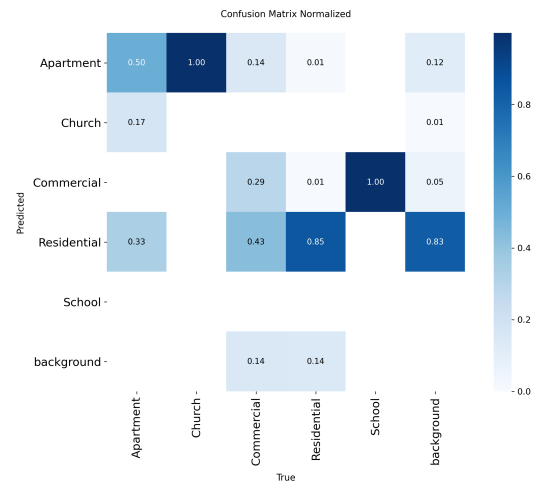
The results of the Logan trained YOLO model are shown in Figure 3. The desired output of the model was to have all the numbers in a diagonal line, which would indicate that the model is predicting the category of a building that is in line with the annotations made.

The model was able to accurately predict residential, commercial, and apartment building types almost 100% of the time. It did underperform with schools and churches. This is likely attributed to the fact that there were very low number of images of both building types provided during model training. Future training would seek out more images of these buildings or possibly find another way to identify them due to their relatively rare occurrence.

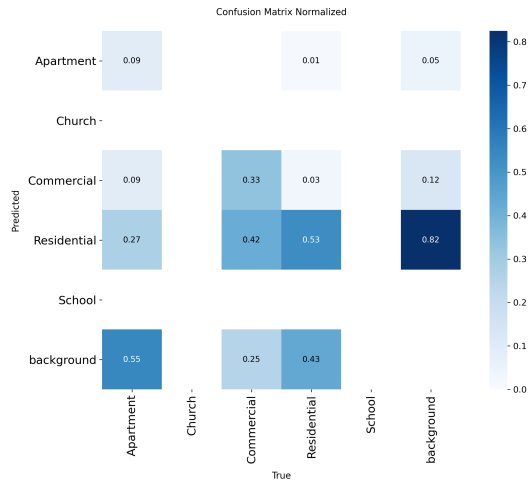
A important aspect to discuss is the issue with the model predicting different building types where there is no building. This is seen on the far right of each graph, where the background is given as the true value, but had predictions in every category. This error occurs from two sources. The first is due to the annotation. Auxiliary building such as backyard sheds were not consistently identified and annotated, but the model still picked them up and registered them. There were



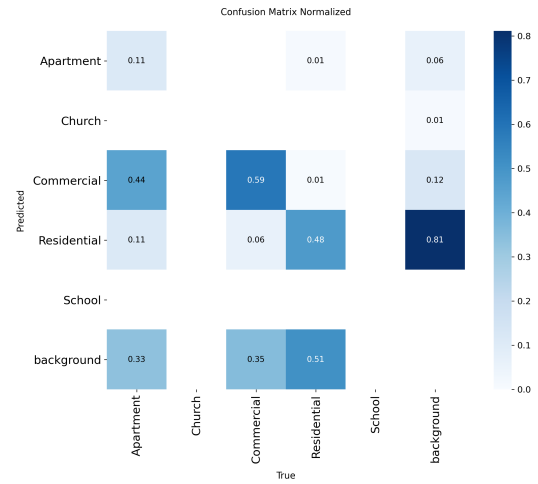
(a) Bountiful, Utah



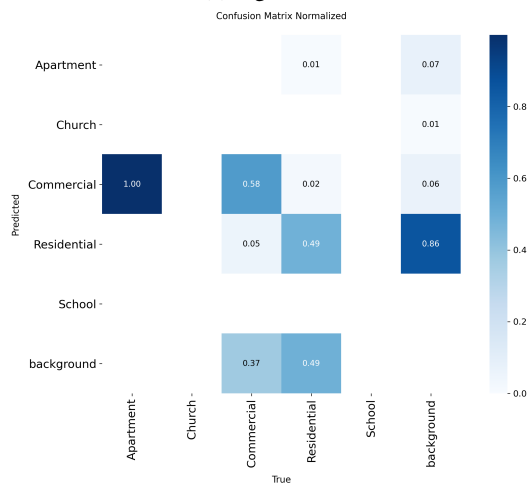
(b) Saratoga Springs, Utah



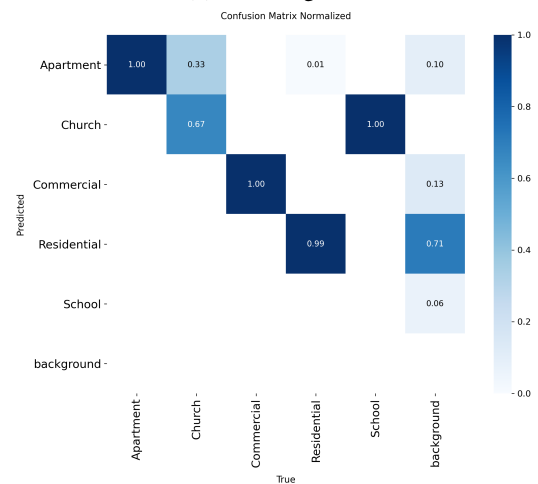
(c) Ogden, Utah



(d) St. George, Utah



(e) Price, Utah



(f) Logan Utah

Figure 4: Normalized YOLO confusion matrixes

also times that only a small portion of a building was present in an image, and it was deemed to be too little of the building for the model to recognize, and thus was not annotated. However, the model in some cases still recognized and categorized the building at a low probability level, while in other cases it did not recognize partial buildings at all. The other source of error was the quality of the photos used. The satellite images from Contextily were convenient and easy to access, but were not high quality. This resulted in the model sometimes interpreting things of solid colors, such as roads and covered parking, as a type of building. The second error could be solved using higher quality photos to help distinguish between these surfaces. The first error could be improved by implementing stricter guidelines on annotation to ensure consistency between data sets. However, in practice, the first error may be less relevant if only the primary building in each image matters to be identified.

### *B. Other Utah Cities Results*

When the Logan model was applied to other cities in Utah, it produced mixed results, as seen in Figure 4. In all cities, the model struggled to correctly identify the different building types. Most of the building annotated and given to the model to identify were residential housing. Yet it misidentified residential building at least half of the time in most cities. Commercial buildings had even worse results, and practically all apartments, schools, and churches, were incorrectly identified. While it seemed to do better in Saratoga Springs with residential buildings, its accuracy with other building types is still far too low to be useful. It is not known why Saratoga Springs residential buildings were so well identified, with the city sharing little with Logan in respect to median income, house age, and location.

The model also struggled with simply recognizing buildings at all. Each city had a significant number of buildings that were predicated as background, meaning the model simply did not pick them up as buildings. When the building images were individually checked, some had trees or other shadows partially covering the roof which could help explain this behavior. Yet, many of the Logan building images have similar characteristics, so this is unlikely to be a factor. The image quality taken from Contextily could have played a role, yet images from both Logan and all other cities seemed to be of similar quality. So, it is uncertain whether higher quality images would have an effect on the model's accuracy in other cities.

While it was theorized that median income or house age would play a significant factor in the model's ability to recognize different building types, the results of the validations did not support this. The model performed roughly the same in every city it was not trained on. Neither house age or house median income seemed to make a difference in the model's ability to classify. While not tested, it also seems that a city's proximity is not a factor, due to all these cities being from Utah. It is possible, however, that cities outside of Utah would

produce even worse results when validated with the Logan model.

Given the results of the validations, a process to develop a model for different locations would be more useful and viable than to create a universal model to be used in a region. It appears that building styles and types are unique enough between cities that a single unified model, at least one mostly dependent on identifying building roofs, is not feasible. However, as will be discussed more in the future research section, adding other environmental variables to the classification, such as the presence of sidewalks, swing sets, etc. could allow for a more general model to be developed.

## IV. CONCLUSION AND FUTURE RESEARCH

This paper presented the results of using a deep learning model to classify building types from aerial imagery. Building footprints in Logan, Utah were used to identify buildings, images of individual buildings were used to train a YOLO model, and that model was then tested on predicated building categories from other cities in Utah in which it was not trained on.

Currently, this research is susceptible to many of the same problems as transportation models. Satellite and aerial imagery are not consistent around the world, having varying availability and image quality. It also may be outdated, especially in fast developing areas. The use of commercial drones could solve many of these problems, but they also come with their own problems. It would take some time to collect a whole city's worth of imagery, the flights would be dependent on the weather, and local and national regulations could present challenges. There are also potential ethical problems with such data collection, with drone collected imagery straddling the line between public and private information. This will need to be considered as further research is done into this area as to stay on the right side of the law [12].

This study is only in the beginning stages of development, with many potential future directions. As discussed previously, a model could be trained to detect other objects that could be used to develop socioeconomic data, such as cars, swing sets, patios, amount of foliage or green space around house, etc. Aerial imagery from drones could also be used to acquire higher quality imagery which could improve the training and accuracy of future models. This imagery could also be used to extract building footprints. The Microsoft Building footprint data set is expansive, but not exhaustive. It also features many notable errors in building footprint identification. Being able to extract more accurate footprints would help with clustering building correctly, which would in turn help with training future models.

Further research could also delve into how other remote sensing data could be utilized and incorporated into building classification. Data such as thermal radiation, nighttime light data, and vegetation index data. Such data could be used to predict statistics like income due to heat use, light use, or amount of vegetation on a property. Or it could give insights

to the number of people living within a household, or whether children are present.

Proximity between buildings is another factor to consider. Similar building types are often next to each other, at least in the United States. Variables could be developed that look at the nearby buildings to the one in question and their predictions for classification. These new variables would influence the ultimate classification of the buildings. These factors could run into issues in areas where mixed use housing is common, or where residential and commercial building are more intertwined.

All these improvements and variables could be used to form a larger model that would take all these variables into account when categorizing buildings. Using methods such as decision trees and random forests, this model would be able to create a process that would not only take YOLO's prediction of the building type into account, but also the other variables mentioned. This encompassing model would seek to not only categorize the buildings but also predict the type of household living in that building.

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